

COMP 2310-Lab 2

1- (8 points) Assume the universe/domain consists of all people.

Translate following statements into logic.

$S(x)$: x is a student

$I(x)$: x is intelligent

$M(x)$: x likes music

(a) all students are intelligent

(b) Some intelligent students like music

(c) Everyone that likes music is not an intelligent student

(d) Only intelligent students like music.

Q 2, Q 3, Q 4: You must write the name of each equivalence or rule that is used.



2- (6 points) Using logical equivalences and inference rules

prove that if $\forall x (P(x) \wedge Q(x))$ then $\forall x P(x) \wedge \forall x Q(x)$.

3- (10 points) Using logical equivalences and inference rules prove

that if $\forall x (L(x) \Rightarrow R(x))$ and $\exists x (P(x) \wedge \sim R(x))$ then

$\sim \forall x (P(x) \Rightarrow L(x))$.

4- (7 points) Given the hypotheses $\exists x (A(x) \wedge \sim B(x))$ and

$\forall x (A(x) \Rightarrow P(x))$ prove that $\exists x (P(x) \wedge \sim B(x))$.